

BE 302: Bioelectric Systems Modeling

Course Code	BE 302
Course Name	Bioelectric Systems Modeling
L-T-P-C	3-0-2-4
Prerequisites	IC136, BE201, IC160
Intended for	B. Tech M.Tech Integrated Dual Degree Bioengineering students
Distribution	Core for IDD BE, Elective for other B.Tech students
Approval	44th BoA (Nov 2021); 37th Senate (Oct 2022), in the B.Tech Bioengineering curriculum

Course Contents

- **Electrical modeling of cells**
Charge transport mechanism in cells, application of circuit theory to cell analysis – electrotonus model, Hodgkin Huxley model for membrane current, voltage changes in cell over space and time. Biological cables – the axons, potential outside a long cylindrical cell, exterior potential for an arbitrary pulse, RC modeling of axon as a transmission line, electrical properties of organs and organ-systems. [14 Lectures]
- **Magnetic modeling of cells**
Magnetic field of a cell in an infinite homogeneous conducting medium, electromagnetic induction, modeling of exterior magnetic field of a cylindrical cell. [2 Lectures]
- **Applications of electrical modeling for biosignal extraction from different organs**
A. *Biosignal extraction from heart*: Origin of cardiac action potential, electric dipole modeling of heart, atrial depolarization causing P wave, sequential ventricular depolarization causing QRS complex, sub-epicardial repolarization causing T wave, recording of electrocardiogram using leads, stimulating the heart – the pacemakers.
B. *Biosignal extraction from brain*: Origin of neuronal action potential, origin of electroencephalogram signals, understanding neural oscillations (Alpha- Beta-, Gamma, Delta- and Theta Waves).
C. *Biosignal extraction at neuromuscular junction*: Origin of action potentials at axon hillock, propagation of bipolar signals in muscle fibers, recording of electromyogram signals.
D. *Detection of weak magnetic fields*: Magnetocardiograms and magnetoencephalograms [8 Lectures]
- **Feedback and control**
Basics of control engineering – notion of open loop and closed loop systems, homeostasis from the electrical model perspective, single loop and multiple loop homeostasis, stability of systems, criteria of stability, example study in neurorehabilitation, closed loop control of blood insulin and glucose regulation, closed loop control of brain stimulation. [18 Lectures]

Laboratory Modules

Laboratory Modules

- Cable model of neurons
- Hodgkin Huxley's model of neurons
- Modeling of exterior electric field of a cylindrical cell
- Modeling of electromagnetic induction in living cells
- Modeling of electrical conduction in heart
- Modeling of electrical conduction in brain
- Modeling of electrical conduction in muscles
- Modeling a feedback loop with one and two time constant(s)
- Modeling of homeostasis process
- Stability analysis of physiological systems

Text Books

1. Eugenio Culurciello, Wei Tang, Evan Joon Park, **Biomedical Circuits and Systems**, CRC Press, 2017.
2. Harold S. Burr, **The Fields of Life. Our Links with the Universe**, Ballantine Publishers, 1973.
3. Robert Berker, Gary Selden, **The Body Electric: Electromagnetism And The Foundation Of Life**, Harper Collins Publishers, 1998.

References

1. K. J. Niklas, **Plant Biomechanics: An engineering approach to plant form and function**, University of Chicago Press, 1992.
2. Ozkaya, Nordin, Goldsheyder and Leger, **Fundamentals of Biomechanics: Equilibrium, Motion, and Deformation**, 3rd Edition, Springer, 2014.
3. R. L. Huston, **Fundamentals of Biomechanics**, CRC Press, 2013.